

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
ProgramName: B. Tech		Assignment Type: Lab	AcademicYear:2025-2026
CourseCoordinatorName		Venkataramana Veeramsetty	
Instructor(s)Name		Dr. V. Venkataramana (Co-ordinator)	
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		Mr. M Prakash	
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		Intern 1 (Dharma teja)	
		Intern 2 (Sai Prasad)	
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		NS_2 (Mounika)	
CourseCode	24CS002PC215	CourseTitle	AI Assisted Coding
Year/Sem	II/I	Regulation	R24
Date and Day of Assignment	Week4 - Tuesday	Time(s)	
Duration	2 Hours	Applicable to Batches	
AssignmentNumber: 7.2(Present assignment number)/24(Total number of assignments)			
Q.No.	Question		Expected Time to complete
1	Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs Lab Objectives: - To identify and correct syntax, logic, and runtime errors in Python programs using AI tools. - To understand common programming bugs and AI-assisted debugging suggestions.		Week4 - Wednesday

- To evaluate how AI explains, detects, and fixes different types of coding errors.
- To build confidence in using AI to perform structured debugging practices.

Lab Outcomes (LOs):

After completing this lab, students will be able to:

- Use AI tools to detect and correct syntax, logic, and runtime errors.
- Interpret AI-suggested bug fixes and explanations.
- Apply systematic debugging strategies supported by AI-generated insights.
- Refactor buggy code using responsible and reliable programming patterns.

Task Description#1

- Task #1 – Syntax Error in Conditionals

```
python

a = 10
if a = 10:
    print("Equal")
```

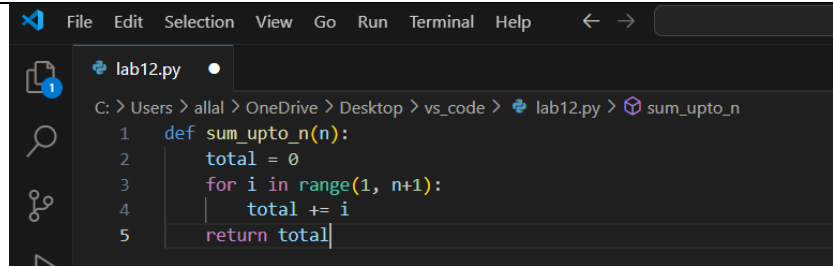
Expected Output#

Task Description#2 (Loops)

- Task #2 – Loop Off-By-One Error.

```
def sum_upto_n(n):
    total = 0
    for i in range(1, n):
        total += i
    return total
```

Expected Output#2



```
File Edit Selection View Go Run Terminal Help
lab12.py
C: > Users > allal > OneDrive > Desktop > vs_code > lab12.py > sum_upto_n
1 def sum_upto_n(n):
2     total = 0
3     for i in range(1, n+1):
4         total += i
5     return total
```

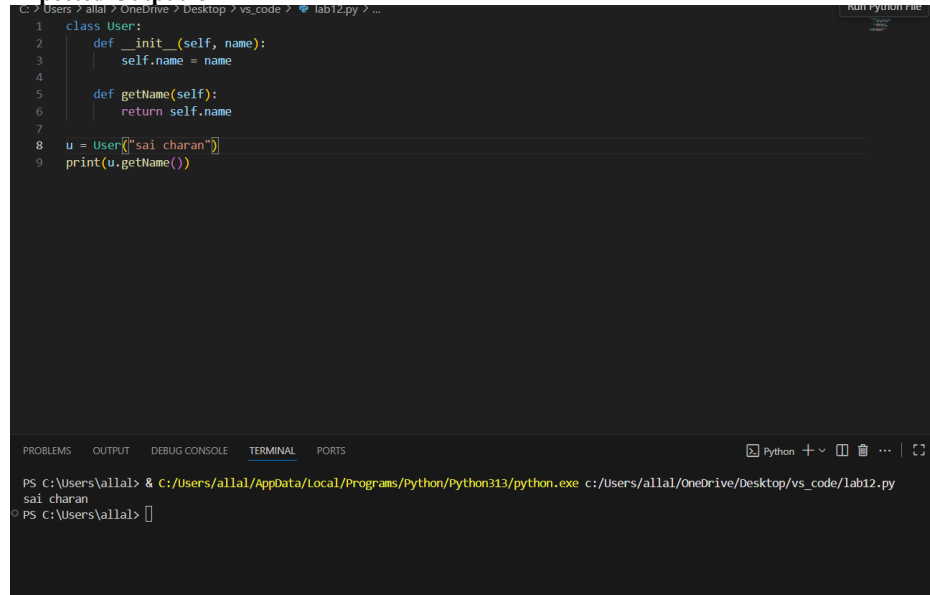
Task Description#3

- Error: AttributeError

```
class User:
    def __init__(self, name):
        self.name = name

u = User("Alice")
print(u.getName())
```

Expected Output#3



```
1 class User:
2     def __init__(self, name):
3         self.name = name
4
5     def getName(self):
6         return self.name
7
8 u = User("sai charan")
9 print(u.getName())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\allal> & c:/Users/allal/AppData/Local/Programs/Python/Python313/python.exe c:/Users/allal/OneDrive/Desktop/vs_code/lab12.py
sai charan
PS C:\Users\allal>

Task Description#4

- Incorrect Class Attribute Initialization

```
class Car:
    def start():
        print("Car started")

mycar = Car()
mycar.start()
```

Expected Output#4

```
lab12.py
C: > Users > allal > OneDrive > Desktop > vs_code > lab12.py > Car > start
1 class Car:
2     def start(self):
3         print("Car not started")
4
5 mycar = Car()
6 mycar.start()

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python + - [ ] [ ] [ ] [ ] [ ]
PS C:\Users\allal> & C:/Users/allal/AppData/Local/Programs/Python/Python313/python.exe c:/Users/allal/OneDrive/Desktop/vs_code/lab12.py
Car not started
PS C:\Users\allal>
```

Task Description#5

- Conditional Logic Error in Grading System

```
def grade_student(score):
    if score < 40:
        return "A"
    elif score < 70:
        return "B"
    else:
        return "C"
```

Expected Output#5

```
lab12.py
C: > Users > allal > OneDrive > Desktop > vs_code > lab12.py > grade_student
1 def grade_student(score):
2     if score >= 70:
3         return "A"
4     elif score >= 40:
5         return "B"
6     else:
7         return "C"
```

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

Evaluation Criteria:

Criteria	Max Marks
Task #1	0.5

	Task #2	0.5		
	Task #3	0.5		
	Task #4	0.5		
	Task #5	0.5		
	Total	2.5 Marks		